

CURRICULUM VITAE

Zakaria Samer Aladem

MECHATRONICS ENGINEER | Robotics

ABOUT ME

Mechatronics Engineer with an interest in robotics and research and development. Strong communication skills, self-motivation and fast learner. Committed to expanding knowledge and contributing to the field.

DETAILS

Amman, Jordan
+962 778144561
zakarialadem2@gmail.com
[LinkedIn](#)

Languages

Arabic- Native Language.
English- Very good at reading, writing, and conversation
IELTS: 6.5

Professional and Academic References

Prof. Dr. Tariq Younes, Mechatronics Engineering Department
Al-Balqa Applied University
tariqmog@bau.edu.jo +962776108619

Prof. Dr. Mohammad Khrisat, Mechatronics Engineering Department
Al-Balqa Applied University
mkhrisat@bau.edu.jo +962775577994

WORK EXPERIENCE

Technical Engineer

Orange Jordan · Internship
Jul 2024 - Oct 2024

In this training, I worked on FDM and SLA 3D printers and laser cutters. My role focused on electronics and waterproofing for an underwater drone. I also gained knowledge in business analysis and management through training sessions.

R&D Engineer

InJo4.0 Jordan · Internship
Jul 2025 – present

Interned at InJo4.0, gaining hands-on experience with Siemens PLCs, Doosan cobots, ESP8266 IoT, and Raspberry Pi embedded systems with computer vision, while strengthening problem-solving and R&D skills through hardware/software integration and experimentation.

Summary of Education

Bachelor's Degree of Mechatronics Engineering, Al-Balqa Applied University, Faculty of Engineering Technology, Amman
2020-2025

Web Site: <http://www.bau.edu.jo>

E-mail: info@fet.edu.jo

GPA: 3.01 **Rating:** Very Good

Projects

Automated Production Line Inspection using Object Detection-(2025)

Developed a real-time quality inspection system for a production line using YOLOv8 object detection, capable of detecting 10 product categories including two defect types. Integrated the detection pipeline with SCADA/HMI via MQTT and Node-RED to automate error detection, trigger corrective actions, and improve production reliability. Optimized performance through dataset balancing, augmentation, and model fine-tuning, achieving robust, high-accuracy inspection in varying conditions.

5-DOF Underwater Drone – (2024)

Developed a remotely operated 5-degree-of-freedom underwater drone equipped with five brushless thrusters, a Raspberry Pi–Pixhawk control system running ArduSub, tethered Ethernet communication, QGroundControl integration, and real-time video streaming with AI-assisted hazard detection. I designed and assembled the waterproof electronics pipe that housed all electronics, created its schematic, and integrated the ESCs, sensors, and communication protocols to ensure stable underwater operation and reliable data transmission.

PET-Machine – Recycling Plastic Bottles into 3D Printer Filament-(2023)

Developed a PET-Machine for recycling plastic bottles into 3D printer filament, implementing a PID-controlled heating system to maintain precise extrusion temperatures. I designed and assembled the electrical systems, created the schematic, and integrated the temperature sensors and control circuits to ensure consistent filament quality and reliable operation.

SKILLS

- **Solidworks** CAD design and simulation
- Embedded Systems using (Arduino-ESP-Raspberry pi)
- MATLAB
- Python
- C++
- Computer Vision
- ROS2
- Electronics design using **LT-Spice**
- Altium
- ROVs

Training Courses

Solid Works Program, Conducted by Moonlight Training Academy, Amman, Jordan
Jun 2022 to Jul 2022

MATLAB Pathway, Conducted by Al Hussein Technical University (HTU) platform HTUx
Feb 2025 to Aug 2025

ROS2, Conducted by Jordan Engineers Association,
OCT 2025 to NOV 2